

LOOK OUT FOR NEW SPIDERS: ARANEAE SPECIES OOD DETECTION USING ENSEMBLE POOLING

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ABSTRACT

The order Araneae, commonly known as spiders, comprises more than 53,000 described species. In the last decade, roughly 9,000 species have been added and discovered, and the pace of discovery shows no signs of slowing down. Species discovery can help in biodiversity evaluation. Existing spider species discovery methods do not commonly incorporate machine learning, which we believe has promising potential to enhance discovery outcomes. To that effect, we present a new method for out-of-distribution (OOD) detection based on Mahalanobis distance scoring using ConvNeXt as a feature embedding model. Our method, OspideR, creates a separate anomaly detection model for each class in the training dataset. At inference time, a separate anomaly score is generated by each model, and then the minimum score is used to determine the final output score of the model, known as the OspideR score. Performance was evaluated using a test set containing in-distribution and out-of-distribution spider images, distinct from the training images. The experimental results show that OspideR achieves better AUROC, F1, and accuracy compared to the other methods tested.

Index Terms— Anomaly detection, OOD detection, machine learning, iNaturalist, spiders.

1. INTRODUCTION

Spiders (Araneae) are among the most prevalent terrestrial predators on Earth. Research on spiders has led to prominent advancements in science and industry, particularly in newly discovered species. A notable example of this is Darwin's bark spider (*Caerostris darwini*). Discovered in 2009, the silk that it produces is extraordinarily tough, being 10 times as tough as Kevlar. It is investigated at a molecular and mechanical level for development of bio-inspired fibers and often serves as the benchmark for toughness [6]. Another example is the K'gari funnel-web spider (*Hadronyche infensa*), whose venom has been shown to protect heart and brain cells from death after heart attacks and strokes. Trials based on its venom are now advancing to human testing [3]. On that basis, our aim is to focus on spiders as a model group for



(a) Darwin's Bark Spider

(b) K'gari Funnel Web Spider

Fig. 1. Images of the *Caerostris darwini* (Darwin's bark spider) and *Hadronyche infensa* (K'gari funnel-web spider).

image-based out-of-distribution (OOD) detection, where the goal is to flag candidate specimens that merit expert review.

In image analysis, anomaly detection (AD) is a specialized field that aims to detect whether a test image deviates significantly from a representation of normality [17]. Supervised AD techniques use training images from one class of normal objects to extract the representation of normality. A commonly used evaluation dataset is an industrial benchmark known as MVTec AD, containing 5354 images and 15 classes, where the anomalies are pixel-level defects [1]. PatchCore is a popular and effective anomaly detection method, which achieves 99.6% image-level AUROC on MVTec AD and is often used as a baseline for evaluation of AD methods.

Within the field of anomaly detection, novelty detection (ND) focuses on detecting images from semantic classes that were not seen during training, rather than detecting images that belong to a known class and also exhibit localized defects [17]. ND methods assign a high anomaly or novelty score to samples that do not align with the "normal" training class. A similar concept is out-of-distribution (OOD) detection, which aims to detect test samples having characteristics outside the distribution of a training dataset without producing a class label [17]. Finally, in deep learning, a related concept is open-set recognition (OSR). Formalized by Scheirer [13], it jointly requires the accurate evaluation of seen classes and unknown classes by appending an ad hoc module onto the tail of existing classifier models [4]. A widely used practical baseline

is the Maximum Logit Score (MLS), which treats the largest pre-softmax activation of a deep learning classifier as a confidence proxy for open set recognition. MLS achieves a 97.9% AUROC score on CIFAR-10 [16].

OOD spider species detection diverges greatly from traditional anomaly detection, which is intended primarily for industrial settings. In MVTec AD, which was originally developed for industrial applications, images within a class exhibit constant pose and background, and abnormalities are localized perturbations. However, in images of spiders in their natural habitat, there is substantial intra-class variation in pose, scale, and background. This causes traditional AD methods to sometimes struggle when applied to natural spider images. In other work, we have seen some more encouraging results for AD/ND methods applied to natural image benchmarks. For instance, Contrastive Learning on Distributionally Shifted Instances (CSI) reports an AUROC score of 93.3% [14] on CIFAR-10 tested against an ImageNet subset. We believe that AD concepts can be applied to spider species detection effectively, despite the challenge of intra-species image variation.

To that effect, for spider species OOD detection, we have developed an OOD detection method named OspideR (Open Spider Recognition). Our method is inspired by the MLS method, which takes the maximum logit score output of a deep learning classifier and uses that value as an effective discriminator between known and unknown images. Similarly, we first generate a group of anomaly scores and then consolidate them to form a single final score, known as an OspideR score in our ensemble pooling method. Instead of using logits from a multi-class model, we construct a separate anomaly detection model for each class in our normal dataset. At test time, for each input image, a pool of anomaly scores equal to the number of models is generated, and the minimum score is output as the representative score of the image. Our method, while rooted in traditional AD concepts, performs significantly better than applying other traditional methods such as PatchCore on natural spider images.

2. DATASET

The datasets for our experiments are sourced from the citizen science platform, iNaturalist [10]. For training OspideR, we curated a subset of 2500 spider images from iNaturalist. We restrict the images to only those released under a Creative Commons license and whose records achieve a Research Grade status in the iNaturalist Data Quality Assessment [10]. This ensures the images are usable for academic purposes and that the quality of the images is sufficient. To ensure adequate sample sizes per species for training and to retain headroom for possible future scaling experiments, we select candidate classes from the top two hundred spider species at iNaturalist in terms of observation count. We arbitrarily designate 25 of these species as “normal”. For OspideR, we construct a training set with 100 images for each of 25 normal classes. Each

class’s dataset is used to train one model in our 25-model evaluation pool, and no images are shared across models. For the ensemble pooling validation (sect. 3.4 below), we further assemble a validation set containing 50 normal and 50 abnormal images. PatchCore, CSI and MLS use 10% of the training set as the validation set. The optimal threshold value is selected as the threshold that maximizes the F1 score on the validation set. A separate testing set containing 50 normal and 50 abnormal images is used for performance evaluation of all the compared methods. All images in the validation and testing sets are distinct from the training set images used for model generation, and the abnormal images are of species that are not from the designated “normal” species. All input images are resized to 224×224 prior to computation. We opt to use a small training dataset for our OOD detector so that it can be instantiated on the fly in the field, where connectivity and computation resources are often limited.

3. METHODS

OspideR (Open Spider Recognition) is an image-level anomaly-detection method for detecting OOD spider images. OspideR aims to separate in-distribution spider images from out-of-distribution images using only normal-class images for training, where the normal dataset includes only spider species that the user defines as normal. Our aim is for our model to be robust to the intra-class variation common in field images, such as background clutter, motion blur, and lighting shifts.

3.1. Feature Extraction and Model Generation

Firstly, we standardize our image inputs to the ConvNeXt feature embedder by resizing and center cropping to a square resolution of 224×224 . Pixel values are normalized channel-wise [8] using the mean and standard deviation of the ImageNet 22k dataset [11]. We use a ConvNeXt-Base encoder pretrained on ImageNet-22k and fine-tuned on ImageNet-1k [8] as the feature extractor; global average pooling of the final stage yields a $d = 1024$ dimensional feature vector. To encourage invariance to orientation [2], we generate four views of each image using label-preserving transforms $\mathcal{T} = \{\text{id}, \text{flip}_h, \text{flip}_v, \text{flip}_{hv}\}$ (identity, horizontal flip, vertical flip, and both). Let $f_\theta(\cdot)$ be the encoder; we obtain per-view features $z_t = f_\theta(t(x))$ for $t \in \mathcal{T}$, L2-normalize each, and aggregate by averaging:

$$z = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{z_t}{\|z_t\|_2}.$$

These view-averaged representations form the normal memory bank. From the bank, we estimate a class-conditional mean μ_c and covariance. Because $N=100 \ll d=1024$, the sample covariance is rank-deficient; we apply the Ledoit–Wolf shrinkage estimator [7] to fit a regularized full covariance matrix Σ_c for each class c .

3.2. Class-Conditional Anomaly Scores

After model generation is completed using the class-conditional training sets, we can obtain per-class anomaly scores for a given input image. For each class, we run the input image through the pretrained ConvNeXt encoder to get its feature embedding for that class. Each class shares the same ConvNeXt encoder. For each class, the image receives an anomaly score based on the Mahalanobis distance [9] to the corresponding multivariate Gaussian model. Concretely, the per-class anomaly score for class c is

$$s_c(x) = \sqrt{(\mathbf{z} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1} (\mathbf{z} - \boldsymbol{\mu}_c)},$$

where $\mathbf{z} \in \mathbb{R}^{1024}$ is the view-averaged embedding of input x , $\boldsymbol{\mu}_c$ is the class-conditional mean, and $\boldsymbol{\Sigma}_c^{-1}$ is the precision matrix obtained from the Ledoit–Wolf estimator. The anomaly score represents the model’s confidence in the image being out-of-distribution — namely, the higher the score, the more likely the image does not belong to that class.

3.3. Ensemble Pooling

For an input image x , we compute the set of per-class anomaly scores $\{s_c(x)\}_{c \in \mathcal{C}}$, which includes one score from each of the 25 class-conditioned models. These anomaly scores are then pooled together, and the minimum is taken to represent the final anomaly score for that test image, known as the OspideR score:

$$s_{\text{OspideR}}(x) = \min_{c \in \mathcal{C}} s_c(x),$$

By selecting the minimum, a single low anomaly score reflects the test image as being recognized as in-distribution by at least one normal model. We apply a validation-calibrated threshold τ^* to $s_{\text{OspideR}}(x)$ to obtain a binary decision for each input image:

$$y_{\text{OspideR}}(x) = \begin{cases} 1 & \text{if } s_{\text{OspideR}}(x) \geq \tau^* \\ 0 & \text{if } s_{\text{OspideR}}(x) < \tau^* \end{cases}$$

The optimal threshold value is determined using our validation dataset.

4. EXPERIMENTAL RESULTS

We compare OspideR against three baselines: PatchCore [12], CSI [14] and Maximum Logit Score (MLS) [16]. PatchCore and MLS use a ConvNeXt-Base backbone, while CSI uses a ResNet18 backbone; CSI does not have ConvNeXt support by default. For PatchCore, we extract features from stages 1 and 2; for hyperparameter values, we set coreset subsampling at 0.15, and num-neighbors equal to 15. For CSI, we use a learning rate of 5×10^{-4} , 4 rotation angles, and train for 5 epochs. For MLS, we train a ConvNeXt classifier

Table 1. Comparison of OspideR to Existing Methods

Method	AUROC	F1	Accuracy
OspideR	89.52	84.48	85.00
PatchCore	51.76	11.32	53.00
CSI	50.40	32.09	45.00
Maximum Logit Score	88.48	78.65	81.00

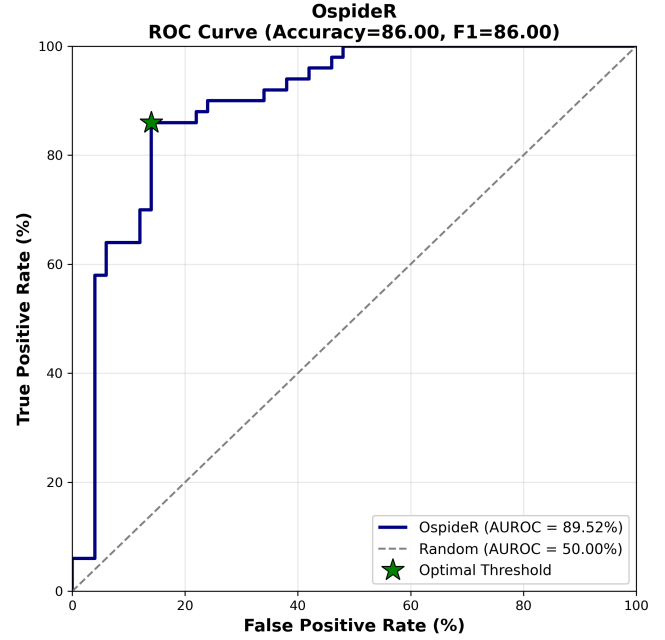


Fig. 2. ROC curve for OspideR applied to the test set.

for 5 epochs with a learning rate of 5×10^{-4} and evaluate using the maximum pre-softmax logit as the score. We report AUROC [5], F1 [15], and accuracy; the aggregate results appear in Table 1. The ROC curve for our method applied to the test set can be seen in Figure 2. The results show that the proposed method achieves better AUROC, F1, and accuracy compared to the other methods tested.

A highlight of OspideR is its ability to ingest new data with minimal retraining. Because we aggregate per-class anomaly scores via min-pooling, adding a new normal species only requires training a new class-conditional model and appending it to the pool. By contrast, in the MLS OSR baseline method, adding classes typically necessitates end-to-end retraining or fine-tuning of the ConvNeXt classifier.

5. CONCLUSION

We have introduced OspideR, an OOD detector for spider species, which uses a ConvNeXt backbone and a Mahalanobis scoring mechanism. The experimental results show that OspideR achieves better AUROC, F1, and accuracy compared to the other methods tested. Further, OspideR is more flexible

than MLS due to its ability to ingest new data with minimal retraining. We believe our work indicates a promising direction for future research in applying anomaly detection techniques to real-world images.

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